

JAVA PRIME PALINDROME PROGRAM

Object-Oriented Programming Practical Record

Question - Sample Java Program to Check Whether the Given Number Is Prime Palindrome or Not (CO1)

Problem Statement: Write a Java program that accepts a number and checks whether it is both prime and palindrome. A prime palindrome is a number that has no factors other than 1 and itself and reads the same forward and backward.

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1. Steps for Execution

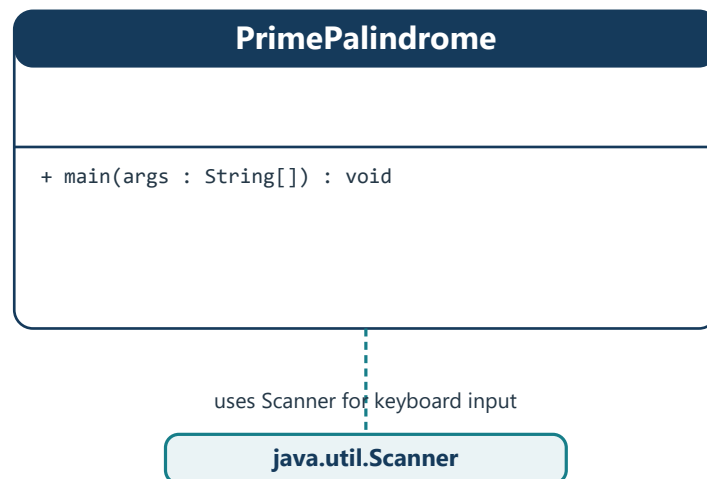
1. Open a Java IDE or command prompt with the Java Development Kit installed.
2. Create a source file named PrimePalindrome.java.
3. Copy the program into the file and save it.
4. Compile it using `javac PrimePalindrome.java`.
5. Run it using `java PrimePalindrome`.
6. Enter an integer when prompted and view the result.

2. Algorithm

1. Start the program and create a Scanner object.
2. Read the number x from the user and copy it into y .
3. Set reverse r to 0. While y is greater than 0, append $y \% 10$ to r and divide y by 10.
4. Set prime to true and palin to false.
5. If r equals x , set palin to true.
6. For every i from 2 through $x / 2$, check whether $x \% i$ is zero. If so, set prime to false and stop the loop.
7. If both prime and palin are true, print that the number is a PrimePalindrome; otherwise print that it is not.
8. Stop.

3. UML Class Diagram

The program contains one main class. It uses Scanner to obtain the input number.



4. Java Source Code

Filename: PrimePalindrome.java. The original program logic is retained; useful comments have been added.

```
import java.util.Scanner;
public class PrimePalindrome {
    public static void main(String[] args) {
        // Create Scanner object to read a number from the user.
        Scanner s = new Scanner(System.in);
        System.out.println("Enter a Number : ");
        int x = s.nextInt();
        // Reverse the given number to check for palindrome.
        int y = x;
        int r = 0;
        while (y > 0) {
            r = (r * 10) + (y % 10);
            y = y / 10;
        }
        boolean prime = true;
        boolean palin = false;
        // A number is a palindrome when it equals its reverse.
        if (r == x)
            palin = true;
        // Check whether the number has a divisor other than 1 and itself.
        for (int i = 2; i <= x / 2; i++) {
            if (x % i == 0) {
                prime = false;
                break;
            }
        }
        // Print the final result only when both conditions are true.
        if (prime && palin)
            System.out.println("Given number is a PrimePalindrome");
        else
            System.out.println("Given number is Not PrimePalindrome");
    }
}
```

5. Representative Sample Output

The first run uses 131, which is prime and reads the same backward. The second run uses 121, which is a palindrome but divisible by 11.

```
Enter a Number :  
131  
Given number is a PrimePalindrome  
Enter a Number :  
121  
Given number is Not PrimePalindrome
```

Result: The Java program was executed successfully and correctly identified whether the supplied number is a prime palindrome.